Solid Geometry & Similarity

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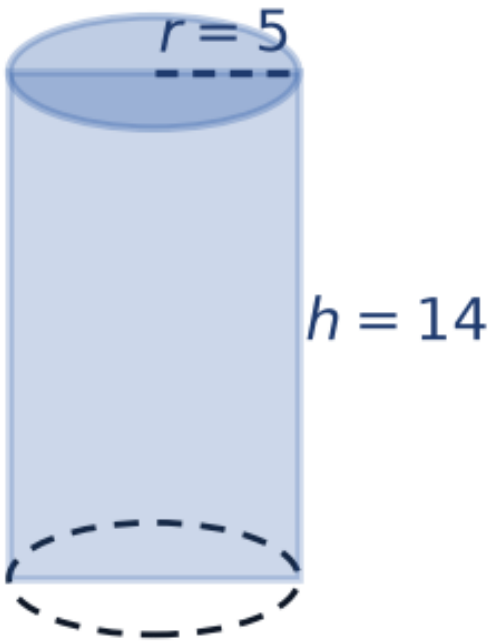
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Volume of prisms and cylinders

Find the volume of a rectangular prism with length 8, width 5, and height 6.

2 [Foundational] • Target: Volume of prisms and cylinders

Find the volume of a cylinder with radius 5 cm and height 14 cm, in terms of π.



3 [Foundational] • Target: Volume of prisms and cylinders

A triangular prism has a triangular cross-section of area 24 cm^2 and length 15 cm. Find its volume.

4 [Foundational] • Target: Volume of prisms and cylinders

Find the volume of a cube with side length 9.

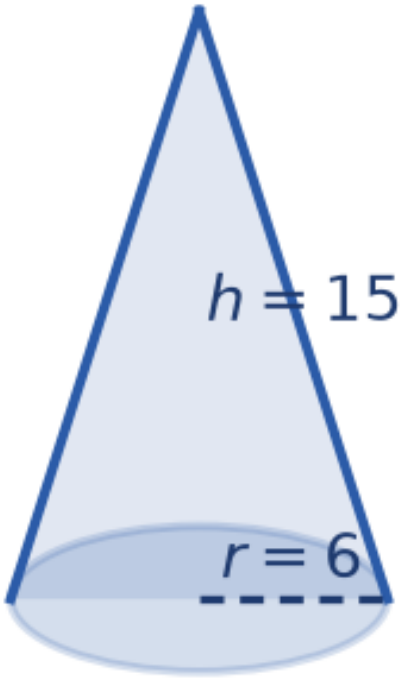
5 [Foundational] • Target: Volume of pyramids and cones

Find the volume of a square pyramid with base side 10 and height 12.

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6 [Foundational] • Target: Volume of pyramids and cones

Find the volume of a cone with radius 6 and height 15, in terms of π.



7 [Foundational] • Target: Volume of pyramids and cones

A cone has volume 300π cubic units and height 25. Find its radius.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Volume of pyramids and cones

A rectangular pyramid has a base 6 by 8 and height 9. Find its volume.

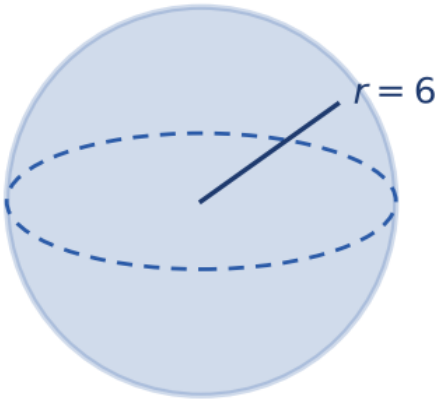
9 [Intermediate] • Target: Volume of spheres

Find the volume of a sphere with radius 9, in terms of π.

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10 [Intermediate] • Target: Volume of spheres

A hemisphere has radius 6. Find its volume, in terms of π.



11 [Intermediate] • Target: Volume of spheres

A spherical tank has diameter 10. Find its volume, in terms of π.

12 [Intermediate] • Target: Volume of spheres

Find the volume and surface area of a sphere with radius 3, in terms of π.

13 [Intermediate] • Target: Surface area of solids

Find the surface area of a cylinder with radius 4 and height 10, in terms of π.

14 [Intermediate] • Target: Surface area of solids

Find the surface area of a cone with radius 5 and slant height 13, in terms of π.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

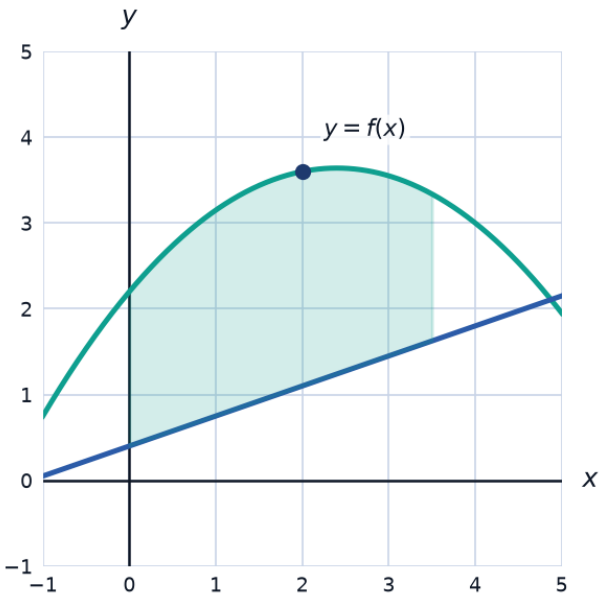
15 [Challenging] • Target: Surface area of solids • Cross-Lesson: 8.01 Statistics & Probability

Find the surface area of a sphere with radius 7, in terms of π.

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16 [Challenging] • Target: Surface area of solids • Cross-Lesson: 1.01 Advanced Algebra

Find the surface area of a cube with side length 7.



17 [Challenging] • Target: Composite solids • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

A solid consists of a cylinder (radius 3, height 10) topped by a cone with the same radius and height 4. Find the total volume, in terms of π.

18 [Challenging] • Target: Composite solids • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

A rectangular box measuring 8 by 6 by 4 has a cylindrical hole (radius 2, drilled fully through the height of 4) removed. Find the remaining volume, in terms of π.

19 [Challenging] • Target: Composite solids • Cross-Lesson: 4.01 Lines, Triangles & Polygons

A solid consists of a cylinder (radius 4, height 8) topped by a hemisphere of the same radius. Find the total volume, in terms of π.

20 [Challenging] • Target: Composite solids • Cross-Lesson: 5.01 Trigonometry

A solid is a rectangular box (10 by 6 by 5) with a square pyramid (base 6 by 6, height 4) carved out of the top. Find the remaining volume.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Volume of prisms and cylinders

Find the volume of a rectangular prism with length 14, width 10, and height 13.

a. 1820

b. 1824

c. 37

d. 1816

22 [Foundational] • Target: Volume of pyramids and cones

Find the volume of a rectangular prism with length 5, width 7, and height 10.

a. 350

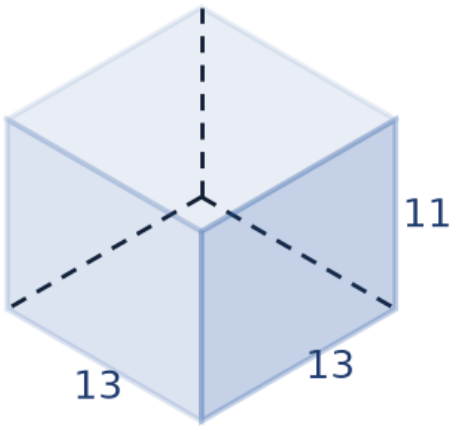
b. 346

c. 354

d. 22

23 [Foundational] • Target: Volume of spheres

Find the volume of a rectangular prism with length 13, width 11, and height 11.



a. 1569

b. 35

c. 1573

d. 1577

24 [Foundational] • Target: Surface area of solids

Find the volume of a rectangular prism with length 8, width 7, and height 13.

a. 28

b. 724

c. 728

d. 732

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25 [Foundational] • Target: Composite solids

Find the volume of a rectangular prism with length 12, width 3, and height 10.

a. 356

b. 364

c. 360

d. 25

26 [Foundational] • Target: Similar solids

Find the volume of a rectangular prism with length 5, width 12, and height 4.

a. 236

b. 240

c. 244

d. 21

27 [Foundational] • Target: Cross-sections of solids

Find the volume of a rectangular prism with length 13, width 15, and height 10.

a. 1946

b. 1954

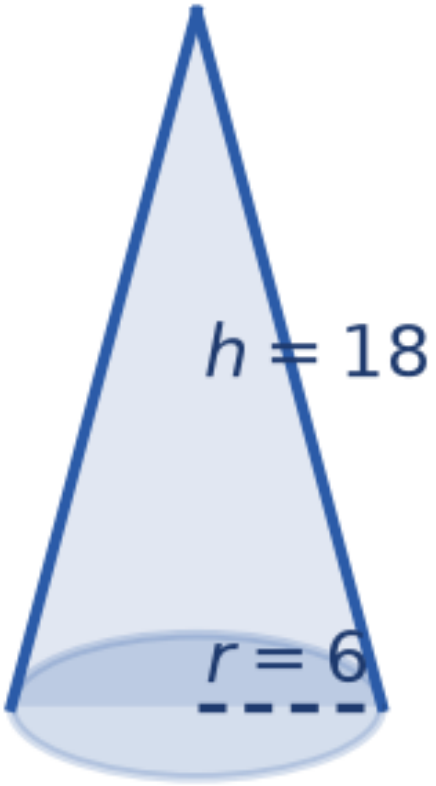
c. 38

d. 1950

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28 [Foundational] • Target: A well-designed word problem: a grain silo

Find the volume of a cone with radius 6 and height 18, in terms of π.



a. 864π

b. 648π

c. 216π

d. 218π

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Polyhedra & Euler's Formula

Find the volume of a cone with radius 3 and height 8, in terms of π.

a. 72π

b. 24π

c. 26π

d. 96π

30 [Intermediate] • Target: Cones, Pyramids & Frustums

Find the volume of a cone with radius 4 and height 9, in terms of π.

a. 48π

b. 50π

c. 192π

d. 144π

31 [Intermediate] • Target: Spheres, Cylinders & Cross-Sections

Find the volume of a cone with radius 3 and height 19, in terms of π.

a. 59π

b. 171π

c. 57π

d. 228π

32 [Intermediate] • Target: Scale Factors & Volume Ratios

Find the volume of a cone with radius 5 and height 14, in terms of π.

a. $\frac{356}{3}\pi$

b. $\frac{350}{3}\pi$

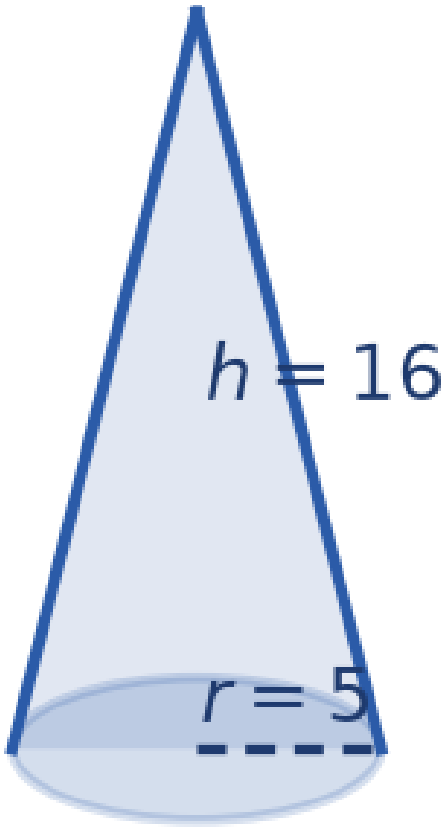
c. 350π

d. $\frac{1400}{3}\pi$

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33 **[Intermediate]** • **Target: Volume of prisms and cylinders**

Find the volume of a cone with radius 5 and height 16, in terms of π.



a. $\frac{1600}{3}\pi$

b. $\frac{406}{3}\pi$

c. $\frac{400}{3}\pi$

d. 400π

34 **[Intermediate]** • **Target: Volume of pyramids and cones**

Find the volume of a cone with radius 8 and height 2, in terms of π.

a. $\frac{128}{3}\pi$

b. 128π

c. $\frac{512}{3}\pi$

d. $\frac{134}{3}\pi$

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35 [Intermediate] • Target: Volume of spheres

Find the volume and surface area of a sphere with radius 7, in terms of π.

a. $\frac{1384}{3}\pi, 196\pi$

b. $\frac{1372}{3}\pi, 204\pi$

c. $\frac{1372}{3}\pi, 196\pi$

d. $\frac{1360}{3}\pi, 188\pi$

36 [Intermediate] • Target: Surface area of solids

Find the volume and surface area of a sphere with radius 11, in terms of π.

a. $\frac{5336}{3}\pi, 484\pi$

b. $\frac{5312}{3}\pi, 476\pi$

c. $\frac{5324}{3}\pi, 484\pi$

d. $\frac{5324}{3}\pi, 492\pi$

37 [Intermediate] • Target: Composite solids

A solid consists of a cylinder (radius 6, height 7) topped by a cone with the same radius and height 7. Find the total volume, in terms of π.

a. 252π

b. 336π

c. 338π

d. 334π

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Similar solids • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

A solid consists of a cylinder (radius 8, height 12) topped by a cone with the same radius and height 2. Find the total volume, in terms of π.

a. $\frac{2438}{3}\pi$

b. 768π

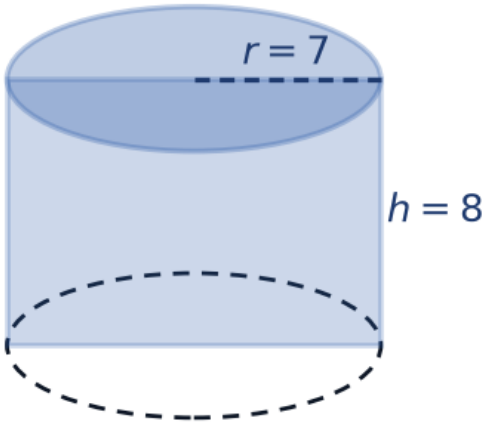
c. $\frac{2432}{3}\pi$

d. $\frac{2426}{3}\pi$

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39 [Challenging] • Target: Cross-sections of solids • Cross-Lesson: 4.01 Lines, Triangles & Polygons

Two similar cylinders have heights 7 and 8. Find the ratio of their volumes.



a. $\frac{512}{343}$

b. $\frac{343}{512}$

c. $\frac{64}{49}$

d. $\frac{8}{7}$

40 [Challenging] • Target: A well-designed word problem: a grain silo • Cross-Lesson: 5.01 Trigonometry

Two similar cylinders have heights 2 and 5. Find the ratio of their volumes.

a. $\frac{8}{125}$

b. $\frac{25}{4}$

c. $\frac{125}{8}$

d. $\frac{5}{2}$

41 [Challenging] • Target: Polyhedra & Euler's Formula • Cross-Lesson: 6.01 Circles & Coordinate Geometry

A convex polyhedron has 12 vertices and 30 edges. How many faces does it have?

a. 18

b. 20

c. 22

d. 24

42 [Challenging] • Target: Polyhedra & Euler's Formula • Cross-Lesson: 8.01 Statistics & Probability

What is the length of the space diagonal of a rectangular prism with dimensions $3, 4,$ and 12?

a. 13

b. 15

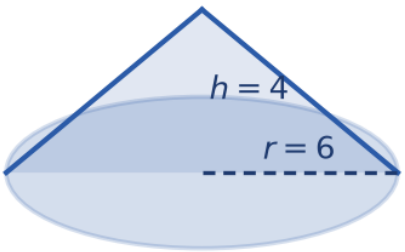
c. $\sqrt{160}$

d. 14

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43 [Challenging] • Target: Cones, Pyramids & Frustums • Cross-Lesson: 1.01 Advanced Algebra

What is the volume of the frustum formed in the above cone with bases of radii 6 cm and 4 cm and height 3 cm?



a. 76π cm^3

b. 72π cm^3

c. 84π cm^3

d. 96π cm^3

44 [Challenging] • Target: Spheres, Cylinders & Cross-Sections • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

A sphere is intersected by a plane 5 cm from its center. If the radius of the cross-sectional circle is 12 cm, what is the radius of the sphere?

a. 13 cm

b. 15 cm

c. $\sqrt{119}$ cm

d. 17 cm

45 [Challenging] • Target: Scale Factors & Volume Ratios • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

Two similar solid statues have heights in the ratio $2:5$. If the smaller statue weighs 16 kg, what does the larger statue weigh (assuming identical material)?

a. 250 kg

b. 100 kg

c. 200 kg

d. 320 kg

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